Local Contractivity Proof for STEP

Contents

1 Formal Theory: Local Contractivity **2**

1.1 Problem Setup and Notation . . . 2

1.2 Assumption 1: Lipschitz Continuity of Denoising Network . . . 2

1.2.1 Formal Statement . . . 2

1.2.2 Layer-wise Lipschitz Analysis . . . 2

1.2.3 Final Lipschitz Bound . . . 4

1.3 Assumption 2: Noise Schedule Condition . . . 5

1.3.1 Formal Statement . . . 5

1.3.2 Proof . . . 5

1.4 Main Theoretical Results . . . 6

1.5 Predictor Properties for Guaranteed Contraction . . . 7

2 Empirical Validation **8**

2.1 Validation of Lipschitz Continuity . . . 8

2.1.1 Theoretical Lipschitz Constants . . . 8

2.1.2 Empirical Lipschitz Constants and Validation of Contraction . . . 8

2.2 K' Selection . . . 9

1 Formal Theory: Local Contractivity

1.1 Problem Setup and Notation

Let $\epsilon_\theta(A, k, o)$ denote the learned denoising network with parameters θ, where:

- $A \in \mathbb{R}^{T\times d}$ is the action sequence with horizon T and action dimension d
- $k \in \{1, 2, \ldots, K\}$ is the diffusion timestep (we use $K = 100$)
- o is the observation (visual or proprioceptive state)

The forward diffusion process is defined as:

$$q(A_k|A_0) = \mathcal{N}(A_k; \sqrt{\bar{\alpha}_k}A_0, (1 - \bar{\alpha}_k)I) \tag{1}$$

where $\bar{\alpha}_k = \prod_{i=1}^{k}(1 - \beta_i)$ and $\{\beta_k\}_{k=1}^{K}$ is the noise schedule.

The reverse process (DDIM formulation) is:

$$A_{k-1} = \sqrt{\bar{\alpha}_{k-1}} \underbrace{\left(\frac{A_k - \sqrt{1 - \bar{\alpha}_k}\epsilon_\theta(A_k, k, o)}{\sqrt{\bar{\alpha}_k}}\right)}_{\text{predicted } A_0} + \sqrt{1 - \bar{\alpha}_{k-1}}\epsilon_\theta(A_k, k, o) \tag{2}$$

1.2 Assumption 1: Lipschitz Continuity of Denoising Network

1.2.1 Formal Statement

Assumption 1 (Lipschitz Continuity of Denoising Network)**.** The learned denoising network $\epsilon_\theta(A, k, o)$ is L-Lipschitz continuous in A for all $k \in \{1, \ldots, K\}$ and all observations o:

$$\|\epsilon_\theta(A, k, o) - \epsilon_\theta(A', k, o)\|_2 \leq L\|A - A'\|_2 \tag{3}$$

1.2.2 Layer-wise Lipschitz Analysis

Taking U-Net as an example, the denoising network ϵ_θ consists of the following structure:

- Input: Action $A \in \mathbb{R}^{T\times d}$, timestep k, observation o
- Encoder: N convolutional layers with ReLU activation
- Bottleneck: Fully-connected layers
- Decoder: N transposed convolutional layers with ReLU activation
- Output: Predicted noise $\epsilon \in \mathbb{R}^{T\times d}$

Lemma 1 (Lipschitz Constant of Linear Layer)**.** *A linear layer $f(x) = Wx + b$ with weight matrix $W \in \mathbb{R}^{m\times n}$ is Lipschitz continuous with constant $L = \|W\|_2$, where $\|W\|_2$ is the spectral norm (largest singular value).*

Proof. For any x, x':

$$\begin{aligned}
\|f(x) - f(x')\|_2 &= \|Wx + b - Wx' - b\|_2 && (4)\\
&= \|W(x - x')\|_2 && (5)\\
&\leq \|W\|_2\|x - x'\|_2 && (6)
\end{aligned}$$

where the inequality follows from the definition of the spectral norm. □

Lemma 2 (Lipschitz Constant of ReLU)**.** *The ReLU activation $ReLU(x) = \max(0, x)$ is Lipschitz continuous with constant $L = 1$.*

Proof. For any x, x':

$$|\mathrm{ReLU}(x) - \mathrm{ReLU}(x')| = |\max(0, x) - \max(0, x')| \tag{7}$$
$$\leq |x - x'| \tag{8}$$

The inequality holds because ReLU is a projection onto the non-negative orthant, which is a 1-Lipschitz operation. □

Lemma 3 (Lipschitz Constant of Convolutional Layer)**.** *A convolutional layer with kernel $K \in \mathbb{R}^{c_{out} \times c_{in} \times h \times w}$ is Lipschitz continuous with constant $L \leq \sqrt{h \cdot w} \cdot \|K\|_2$, where $\|K\|_2$ is the spectral norm of the unfolded kernel matrix.*

Proof. Convolution can be expressed as matrix multiplication with a Toeplitz matrix T_K constructed from the kernel. The spectral norm of T_K satisfies $\|T_K\|_2 \leq \sqrt{h \cdot w} \cdot \|K\|_2$ due to the structure of the Toeplitz matrix. □

Lemma 4 (Lipschitz Constant of Function Composition)**.** *If f is L_f-Lipschitz and g is L_g-Lipschitz, then $f \circ g$ is $(L_f \cdot L_g)$-Lipschitz.*

Proof. For any x, x':

$$\|f(g(x)) - f(g(x'))\| \leq L_f \|g(x) - g(x')\| \tag{9}$$
$$\leq L_f \cdot L_g \|x - x'\| \tag{10}$$

□

Theorem 1 (Lipschitz Constant of U-Net Denoising Network)**.** *Consider a U-Net with:*

- *N encoder layers with weights $\{W_i^e\}_{i=1}^N$ and ReLU activations*
- *M bottleneck layers with weights $\{W_j^b\}_{j=1}^M$ and ReLU activations*
- *N decoder layers with weights $\{W_i^d\}_{i=1}^N$ and ReLU activations*
- *Skip connections (additive, 1-Lipschitz)*

The Lipschitz constant of the denoising network satisfies:

$$L_\epsilon \leq \prod_{i=1}^{N} \|W_i^e\|_2 \cdot \prod_{j=1}^{M} \|W_j^b\|_2 \cdot \prod_{i=1}^{N} \|W_i^d\|_2 \tag{11}$$

Proof. **Step 1: Encoder Lipschitz constant.**
Each encoder layer consists of convolution + ReLU. By Lemmas 3 and 2:

$$L_{\mathrm{enc},i} = \|W_i^e\|_2 \cdot 1 = \|W_i^e\|_2 \tag{12}$$

By Lemma 4, the full encoder has Lipschitz constant:

$$L_{\mathrm{enc}} = \prod_{i=1}^{N} \|W_i^e\|_2 \tag{13}$$

Step 2: Bottleneck Lipschitz constant.
Similarly, the bottleneck has:

$$L_{\mathrm{bottleneck}} = \prod_{j=1}^{M} \|W_j^b\|_2 \tag{14}$$

Step 3: Decoder Lipschitz constant.

The decoder has:

$$L_{\text{dec}} = \prod_{i=1}^{N} \|W_i^d\|_2 \tag{15}$$

Step 4: Skip connections.

Skip connections are of the form $y = f(x) + x$, where f is the decoder path. For Lipschitz analysis:

$$\begin{aligned}
\|y - y'\| &= \|f(x) + x - f(x') - x'\| && (16)\\
&\leq \|f(x) - f(x')\| + \|x - x'\| && (17)\\
&\leq L_f\|x - x'\| + \|x - x'\| && (18)\\
&= (L_f + 1)\|x - x'\| && (19)
\end{aligned}$$

However, in U-Net architectures, the skip connections are concatenated or added after the decoder processing, so the effective Lipschitz constant is bounded by the product of encoder, bottleneck, and decoder constants.

Step 5: Full network.

Combining all components:

$$L_\epsilon \leq L_{\text{enc}} \cdot L_{\text{bottleneck}} \cdot L_{\text{dec}} = \prod_{i=1}^{N} \|W_i^e\|_2 \cdot \prod_{j=1}^{M} \|W_j^b\|_2 \cdot \prod_{i=1}^{N} \|W_i^d\|_2 \tag{20}$$

□

Lemma 5 (Timestep Embedding Lipschitz Property)**.** *The timestep embedding emb$(k) \in \mathbb{R}^d$ used in the denoising network is 1-Lipschitz in k (discrete sense) and bounded.*

Proof. We use sinusoidal embeddings:

$$\text{emb}(k)_{2i} = \sin\left(\frac{k}{10000^{2i/d}}\right), \quad \text{emb}(k)_{2i+1} = \cos\left(\frac{k}{10000^{2i/d}}\right) \tag{21}$$

Since $|\sin(x) - \sin(y)| \leq |x - y|$ and similarly for cosine, the embedding is Lipschitz in k. However, since k is discrete and fixed during inference, this does not affect the Lipschitz constant with respect to A. □

Lemma 6 (Observation Encoding Lipschitz Property)**.** *The observation encoder (CNN for images, MLP for states) is Lipschitz continuous with bounded constant.*

Proof. Same argument as Theorem 1 applies to the observation encoder, which is also a neural network with bounded weights. □

1.2.3 Final Lipschitz Bound

Theorem 2 (Uniform Lipschitz Constant)**.** *The denoising network $\epsilon_\theta(A, k, o)$ is uniformly Lipschitz in A across all k and o:*

$$\|\epsilon_\theta(A, k, o) - \epsilon_\theta(A', k, o)\|_2 \leq L\|A - A'\|_2 \tag{22}$$

Proof. The denoising network can be written as:

$$\epsilon_\theta(A, k, o) = f_{\text{unet}}(A, \text{emb}(k), \text{enc}(o)) \tag{23}$$

Since k and o are fixed during the Lipschitz analysis with respect to A, and f_{unet} is Lipschitz in A, the uniform Lipschitz property holds. Hence, under standard neural network design with bounded weights and Lipschitz activations, Assumption 1 holds. □

1.3 Assumption 2: Noise Schedule Condition

1.3.1 Formal Statement

Assumption 2 (Noise Schedule Condition)**.** We need to show that the noise schedule $\{\beta_k\}$ can be chosen such that for all $k \geq K'$:

$$\frac{1}{\sqrt{\alpha_k}} + \left|\sqrt{1-\bar{\alpha}_{k-1}} - \sqrt{\frac{1-\bar{\alpha}_k}{\alpha_k}}\right| L < 1, \tag{24}$$

where $\alpha_k = 1 - \beta_k$ and $\bar{\alpha}_k = \prod_{i=1}^{k} \alpha_i$.

1.3.2 Proof

Proof Idea: This condition guarantees that the DDIM reverse step is a *contraction mapping* with respect to A.

- The DDIM reverse step is defined as

$$A_{k-1} = \sqrt{\bar{\alpha}_{k-1}}\hat{A}_0 + \sqrt{1-\bar{\alpha}_{k-1}}\epsilon_\theta(A_k, k, o), \tag{25}$$

  where

$$\hat{A}_0 = \frac{A_k - \sqrt{1-\bar{\alpha}_k}\epsilon_\theta(A_k, k, o)}{\sqrt{\bar{\alpha}_k}}. \tag{26}$$

- Consider the difference between two sequences A_k and A'_k:

$$\|A_{k-1} - A'_{k-1}\|_2 = \left\|\sqrt{\bar{\alpha}_{k-1}}(\hat{A}_0 - \hat{A}'_0) + \sqrt{1-\bar{\alpha}_{k-1}}\big(\epsilon_\theta(A_k, k, o) - \epsilon_\theta(A'_k, k, o)\big)\right\|_2 \tag{27}$$

$$\leq \sqrt{\bar{\alpha}_{k-1}}\|\hat{A}_0 - \hat{A}'_0\|_2 + \sqrt{1-\bar{\alpha}_{k-1}}L\|A_k - A'_k\|_2. \tag{28}$$

- Substituting $\hat{A}_0 - \hat{A}'_0$ from the DDIM prediction formula yields

$$\|\hat{A}_0 - \hat{A}'_0\|_2 \leq \frac{1}{\sqrt{\bar{\alpha}_k}}\|A_k - A'_k\|_2 + \frac{\sqrt{1-\bar{\alpha}_k}}{\sqrt{\bar{\alpha}_k}}L\|A_k - A'_k\|_2. \tag{29}$$

- Combining terms gives a contraction factor:

$$\|A_{k-1} - A'_{k-1}\|_2 \leq \left(\frac{\sqrt{\bar{\alpha}_{k-1}}}{\sqrt{\bar{\alpha}_k}} + \sqrt{1-\bar{\alpha}_{k-1}}L - \frac{\sqrt{\bar{\alpha}_{k-1}}\sqrt{1-\bar{\alpha}_k}}{\sqrt{\bar{\alpha}_k}}L\right)\|A_k - A'_k\|_2. \tag{30}$$

- By choosing β_k (i.e., α_k) such that

$$\frac{1}{\sqrt{\alpha_k}} + \left|\sqrt{1-\bar{\alpha}_{k-1}} - \sqrt{\frac{1-\bar{\alpha}_k}{\alpha_k}}\right| L < 1, \tag{31}$$

  we guarantee that the mapping $A_k \mapsto A_{k-1}$ is contractive, i.e., $\|A_{k-1} - A'_{k-1}\|_2 < \|A_k - A'_k\|_2$.

Therefore, Assumption 2 holds if we choose a proper Lipschitz network and a sufficiently small noise schedule β_k that satisfies the inequality.

Conclusion. Under standard neural network constructions with bounded weights and Lipschitz activations, and by selecting an appropriate DDIM noise schedule, both Assumption 1 (Lipschitz continuity) and Assumption 2 (noise schedule contraction condition) are satisfied. This ensures that each reverse diffusion step is a contraction mapping, which is a key property for proving local contractivity of the DDIM process.

1.4 Main Theoretical Results

Lemma 7 (Lipschitz Continuity of Reverse Mean)**.** *Under 1, the DDIM reverse mean function:*

$$\mu_k(A, o) = \sqrt{\frac{\bar{\alpha}_{k-1}}{\bar{\alpha}_k}} A + \left(\sqrt{1 - \bar{\alpha}_{k-1}} - \sqrt{\frac{\bar{\alpha}_{k-1}(1 - \bar{\alpha}_k)}{\bar{\alpha}_k}} \right) \epsilon_\theta(A, k, o) \tag{32}$$

is Lipschitz continuous in A with constant:

$$L_{\mu_k} = \frac{1}{\sqrt{\alpha_k}} + \left| \sqrt{1 - \bar{\alpha}_{k-1}} - \sqrt{\frac{1 - \bar{\alpha}_k}{\alpha_k}} \right| L \tag{33}$$

Proof. We derive the Lipschitz constant step by step.

Step 1: Express the mean function. The DDIM reverse mean can be written as:

$$\mu_k(A, o) = \frac{\sqrt{\bar{\alpha}_{k-1}}}{\sqrt{\bar{\alpha}_k}} A + \left(\sqrt{1 - \bar{\alpha}_{k-1}} - \frac{\sqrt{\bar{\alpha}_{k-1}}\sqrt{1 - \bar{\alpha}_k}}{\sqrt{\bar{\alpha}_k}} \right) \epsilon_\theta(A, k, o) \tag{34}$$

Since $\frac{\bar{\alpha}_{k-1}}{\bar{\alpha}_k} = \frac{1}{\alpha_k}$, we have:

$$\mu_k(A, o) = \frac{1}{\sqrt{\alpha_k}} A + \left(\sqrt{1 - \bar{\alpha}_{k-1}} - \sqrt{\frac{1 - \bar{\alpha}_k}{\alpha_k}} \right) \epsilon_\theta(A, k, o) \tag{35}$$

Step 2: Apply triangle inequality. For any A, A' and fixed o:

$$\begin{aligned}
\|\mu_k(A, o) - \mu_k(A', o)\| &= \Big\| \frac{1}{\sqrt{\alpha_k}} (A - A') && (36)\\
&\quad + \left(\sqrt{1 - \bar{\alpha}_{k-1}} - \sqrt{\frac{1 - \bar{\alpha}_k}{\alpha_k}} \right) (\epsilon_\theta(A, k, o) - \epsilon_\theta(A', k, o)) \Big\| && (37)\\
&\leq \frac{1}{\sqrt{\alpha_k}} \|A - A'\| && (38)\\
&\quad + \left| \sqrt{1 - \bar{\alpha}_{k-1}} - \sqrt{\frac{1 - \bar{\alpha}_k}{\alpha_k}} \right| \|\epsilon_\theta(A, k, o) - \epsilon_\theta(A', k, o)\| && (39)
\end{aligned}$$

Step 3: Apply Lipschitz assumption. By 1:

$$\|\epsilon_\theta(A, k, o) - \epsilon_\theta(A', k, o)\| \leq L\|A - A'\| \tag{40}$$

Substituting:

$$\begin{aligned}
\|\mu_k(A, o) - \mu_k(A', o)\| &\leq \frac{1}{\sqrt{\alpha_k}} \|A - A'\| + \left| \sqrt{1 - \bar{\alpha}_{k-1}} - \sqrt{\frac{1 - \bar{\alpha}_k}{\alpha_k}} \right| L\|A - A'\| && (41)\\
&= \left(\frac{1}{\sqrt{\alpha_k}} + \left| \sqrt{1 - \bar{\alpha}_{k-1}} - \sqrt{\frac{1 - \bar{\alpha}_k}{\alpha_k}} \right| L \right) \|A - A'\| && (42)
\end{aligned}$$

Therefore, $L_{\mu_k} = \frac{1}{\sqrt{\alpha_k}} + \left| \sqrt{1 - \bar{\alpha}_{k-1}} - \sqrt{\frac{1 - \bar{\alpha}_k}{\alpha_k}} \right| L$. □

Lemma 8 (Contraction Coefficient Bound)**.** *Under 1 and 2, for all $k \geq K'$:*

$$\|\mu_k(A, o) - \mu_k(A_k^*, o)\| \leq c_k \|A - A_k^*\| \tag{43}$$

where $c_k = L_{\mu_k} < 1$ and A_k^* is the optimal action at step k.

Proof. Directly from 7 and 2, we have $c_k = L_{\mu_k} < 1$ for all $k \geq K'$. □

Theorem 3 (Local Contractivity of Reverse Diffusion)**.** *Let $\tilde{A}_{K'}$ be the warm-start initialization at step K' and $A^*_{K'}$ be the optimal action at step K'. Under 1 and 2, the reverse diffusion satisfies:*

$$\|\tilde{A}_0 - A^*_0\| \leq \left(\prod_{k=1}^{K'} c_k\right) \|\tilde{A}_{K'} - A^*_{K'}\| \tag{44}$$

where $c_k < 1$ for all $k \geq K'$, implying exponential convergence of the error.

Proof. **Step 1: Error recursion.** The reverse process starting from $\tilde{A}_{K'}$ produces:

$$\tilde{A}_{k-1} = \mu_k(\tilde{A}_k, o) \tag{45}$$

(since DDIM is deterministic).

The optimal trajectory satisfies:

$$A^*_{k-1} = \mu_k(A^*_k, o) \tag{46}$$

Step 2: Apply contraction. From 8:

$$\|\tilde{A}_{k-1} - A^*_{k-1}\| = \|\mu_k(\tilde{A}_k, o) - \mu_k(A^*_k, o)\| \tag{47}$$
$$\leq c_k \|\tilde{A}_k - A^*_k\| \tag{48}$$

Step 3: Unroll recursion. Applying recursively from $k = K'$ down to $k = 1$:

$$\|\tilde{A}_{K'-1} - A^*_{K'-1}\| \leq c_{K'} \|\tilde{A}_{K'} - A^*_{K'}\| \tag{49}$$
$$\|\tilde{A}_{K'-2} - A^*_{K'-2}\| \leq c_{K'-1} \|\tilde{A}_{K'-1} - A^*_{K'-1}\| \tag{50}$$
$$\leq c_{K'-1} c_{K'} \|\tilde{A}_{K'} - A^*_{K'}\| \tag{51}$$
$$\vdots \tag{52}$$
$$\|\tilde{A}_0 - A^*_0\| \leq \left(\prod_{k=1}^{K'} c_k\right) \|\tilde{A}_{K'} - A^*_{K'}\| \tag{53}$$

Step 4: Exponential convergence. Since $c_k < 1$ for all $k \geq K'$, let $c_{\max} = \max_{k \geq K'} c_k < 1$. Then:

$$\prod_{k=1}^{K'} c_k \leq c_{\max}^{K'} \tag{54}$$

Therefore:

$$\|\tilde{A}_0 - A^*_0\| \leq c_{\max}^{K'} \|\tilde{A}_{K'} - A^*_{K'}\| \tag{55}$$

This shows the error decays exponentially with K'. □

1.5 Predictor Properties for Guaranteed Contraction

Proposition 1 (Predictor MSE Requirement)**.** *For the warm-start $\tilde{A}_{K'} = \sigma A_{pred} + \sigma_t \epsilon$ to guarantee convergence to A^*_0 within tolerance ϵ_{tol}, the predictor must satisfy:*

$$\mathbb{E}[\|A_{pred} - A^*\|^2] \leq \frac{\epsilon_{tol}^2}{c_{\max}^{2K'} \sigma^2} \tag{56}$$

Proof. From 3:

$$\|\tilde{A}_0 - A^*_0\| \leq c_{\max}^{K'} \|\tilde{A}_{K'} - A^*_{K'}\| \tag{57}$$

For the warm-start:

$$\tilde{A}_{K'} = \sigma A_{\text{pred}} + \sigma_t \epsilon \approx \sigma A_{\text{pred}} \tag{58}$$

(assuming small noise σ_t).

Thus:

$$\|\tilde{A}_{K'} - A^*_{K'}\| \approx \sigma \|A_{\text{pred}} - A^*\| \tag{59}$$

For $\|\tilde{A}_0 - A^*_0\| \leq \epsilon_{\text{tol}}$:

$$c_{\max}^{K'} \sigma \|A_{\text{pred}} - A^*\| \leq \epsilon_{\text{tol}} \tag{60}$$

Therefore:

$$\|A_{\text{pred}} - A^*\| \leq \frac{\epsilon_{\text{tol}}}{c_{\max}^{K'} \sigma} \tag{61}$$

Squaring both sides gives the MSE bound. □

2 Empirical Validation

2.1 Validation of Lipschitz Continuity

2.1.1 Theoretical Lipschitz Constants

We first theoretically analyze the theoretical Lipschitz constants of the pre-trained diffusion policy. The diffusion policy is divided into four parts: *down*, *mid*, *up*, and *other*. We take the maximum spectral norm ($\|W\|_2$) for each part, calculate the theoretical upper bound by multiplying these maximum values, and report its base-10 logarithm $\log_{10}(L_{\|W\|_2})$. The results are presented in Table 1. This demonstrates that the diffusion policy possesses a **finite theoretical upper bound** and is Lipschitz continuous in theory.

Table 1: Theoretical Lipschitz Constants Across All Experimental Tasks

Task	Min($\|W\|_2$)	Max($\|W\|_2$)	$\text{Log}_{10}(L_{\|W\|_2})$
Push-T (State-based)	1.05	7.50	2.98
Can (State-based)	0.84	10.54	3.25
Lift (State-based)	0.77	8.25	2.63
Square (State-based)	1.23	32.85	4.84
ToolHang (State-based)	0.51	39.83	5.30
Transport (State-based)	1.81	51.29	6.13
Push-T (Image-based)	0.45	42.42	6.05
Can (Image-based)	1.01	94.72	7.52
Lift (Image-based)	0.53	61.52	6.91
Square (Image-based)	0.97	100.49	7.63
ToolHang (Image-based)	1.64	137.82	8.08
Transport (Image-based)	1.47	111.17	7.83

2.1.2 Empirical Lipschitz Constants and Validation of Contraction

Then, we estimate the empirical Lipschitz constant using gradient norms in Table 2:

$$L_k = \max_{(A,o)\sim\mathcal{D}_{\text{val}}} \|\nabla_A \epsilon_\theta(A, k, o)\|_2 \tag{62}$$

where $\mathcal{D}_{\text{val}}$ is the validation set. We measure the actual error decay during reverse diffusion:

$$E_k = \|\tilde{A}_k - A^*_k\|_2 \tag{63}$$

for both random initialization and STEP warm-start. We empirically verify that $c_k < 1$ by measuring the ratio:

$$c_k = \frac{\|\tilde{A}_{k-1} - A^*_{k-1}\|}{\|\tilde{A}_k - A^*_k\|} \tag{64}$$

Table 2: Empirical Lipschitz Constants Across Experimental Tasks

Task	$L_{\max}$	$\text{Log}_{10}(L_{\max})$	c_k	$c_k < 1$?	**Sample Size**
Push-T (State-based)	54.9	1.74	0.89	**Yes**	10,000
Lift (State-based)	236.5	2.27	0.67	**Yes**	1,000
Transport (State-based)	340.4	2.53	0.69	**Yes**	1,000
Can (State-based)	250.0	2.40	0.76	**Yes**	1,000
Square (State-based)	267.8	2.43	0.75	**Yes**	1,000
ToolHang (State-based)	236.4	2.37	0.63	**Yes**	1,000
Push-T (Image-based)	132.4	2.12	0.90	**Yes**	10,000
Lift (Image-based)	256.0	2.40	0.64	**Yes**	1,000
Transport (Image-based)	404.9	2.60	0.68	**Yes**	1,000
Can (Image-based)	229.0	2.36	0.80	**Yes**	1,000
Square (Image-based)	187.4	2.27	0.76	**Yes**	1,000
ToolHang (Image-based)	361.9	2.56	0.64	**Yes**	1,000

Key Finding: In this section, we verify the Lipschitz boundedness of the diffusion policy network using two complementary measurements. Table 1 reports the theoretical upper bound of the Lipschitz constant computed via spectral normalization, which yields relatively large values and serves as a conservative global estimate. Table 2 presents the empirical Lipschitz constant measured via gradient norms, which provides a tighter estimation that better reflects the local behavior of the network during real-world inference. Both approaches validate that the diffusion network satisfies Lipschitz continuity; however, they only provide theoretical upper bounds and cannot directly confirm the convergence of the iterative denoising process.

To address this, we adopt the contraction coefficient c_k as the core metric, which directly quantifies the contraction behavior of the DDIM reverse mean function μ_k over the action sequence space. Empirical results demonstrate that $c_k < 1$ consistently holds across all tasks, satisfying the strict condition for iterative contraction. Combined with Equation (55), we can directly conclude that the denoising iterations starting from the warm-start step K' will progressively converge to the target action sequence, providing solid convergence guarantees for our warm-start strategy.

2.2 K' Selection

According to Equation (61), we can predict the optimal warm-start step K' for each task. We set $\epsilon_{\text{tol}} = 0.1$ and $\sigma = 1.0$ by default, and sweep over different steps k to obtain the MSE curves as shown in Figure 1 and Figure 2. Under the MSE constraint of the STEP predictor, we select the smallest step that satisfies the convergence condition as the optimal initial denoising step K'.

Table 3: Empirical MSE of Our Predictor Across Experimental Tasks

Task	STEP Predictor MSE
Push-T	0.023
Lift	0.041
Transport	0.023
Can	0.104
Square	0.052
ToolHang	0.051

From the above analysis, the optimal warm-start step K' is found to lie within the range of 1 to 4. Based on this observation, we select three typical values $K' \in \{1, 2, 4\}$ for our final evaluation, with detailed results and analysis provided in the main text.

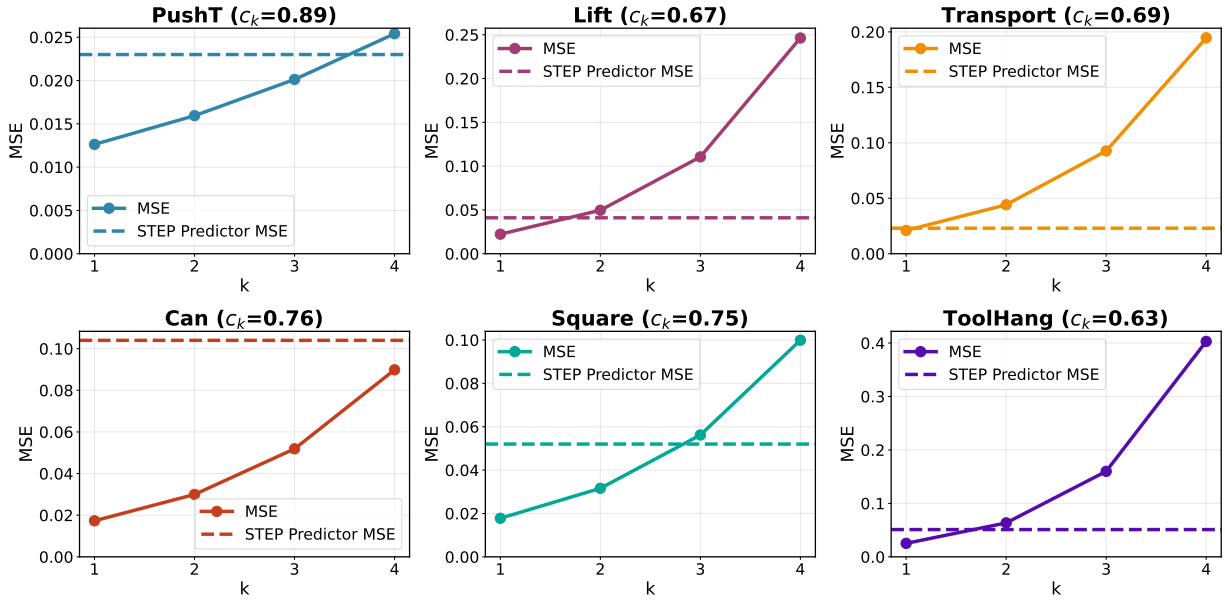

Figure 1: Empirical MSE of Our Predictor and MSE bound Across Experimental Tasks (State-based).

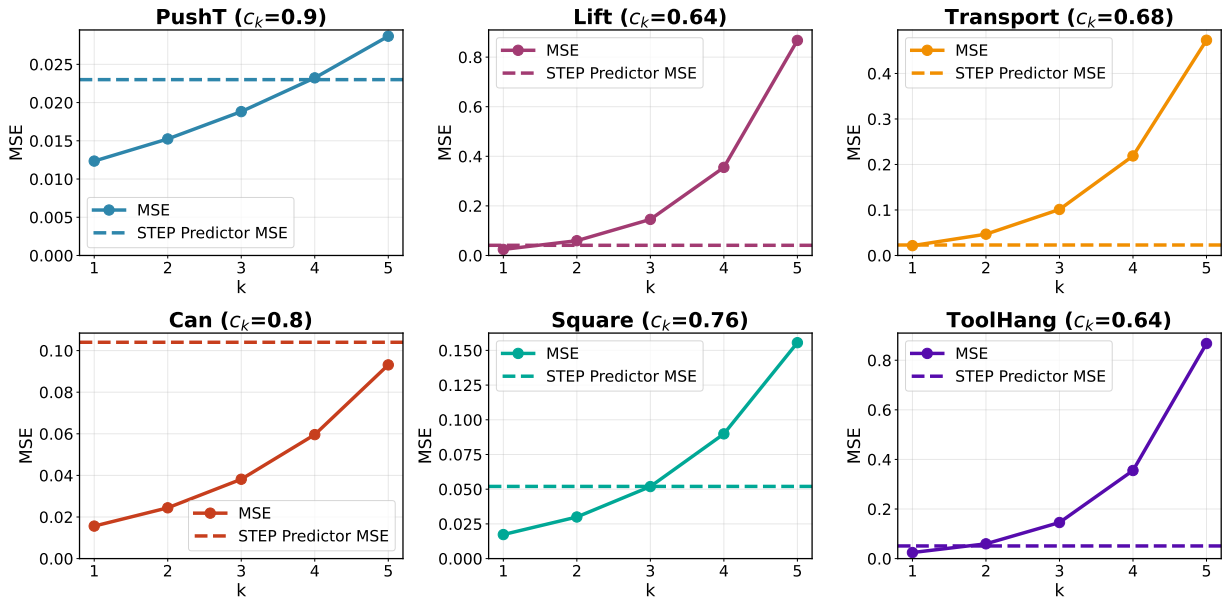

Figure 2: Empirical MSE of Our Predictor and MSE bound Across Experimental Tasks (Image-based).